



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<https://github.com/farturo93/applied-data-science-capstone.git>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Collected SpaceX Falcon 9 launch data using the SpaceX REST API and complementary web scraping techniques to build a comprehensive launch dataset.
- Performed data wrangling to clean, standardize, and prepare the data for exploratory analysis and predictive modeling.
- Conducted exploratory data analysis (EDA) using data visualization and SQL queries to identify relationships among launch sites, payload mass, orbit types, booster versions, and mission outcomes.
- Developed interactive analytics with Folium maps and a Plotly Dash dashboard to visualize launch locations, mission success rates, and payload trends.
- Built and evaluated multiple machine learning classification models to predict Falcon 9 first-stage landing success using historical launch characteristics.
- Compared model performance using classification metrics and selected the best-performing model based on predictive accuracy.
- The project demonstrates how data collection, exploratory analysis, interactive visualization, and machine learning can be combined to generate insights into SpaceX launch operations and support prediction of reusable booster landing success.

Introduction

- SpaceX has significantly reduced launch costs through reusable Falcon 9 first-stage boosters.
- Predicting first-stage landing success is an important step in assessing mission outcomes and operational efficiency.
- This project applies data science techniques to analyze historical Falcon 9 launch data and develop predictive models.
- The analysis integrates data collection, exploratory data analysis, interactive visualization, and machine learning into a complete analytical workflow.

Project Background

- Historical Falcon 9 launch records were collected from publicly available data sources.
- The dataset includes launch characteristics such as launch site, payload mass, orbit type, booster version, and landing outcome.
- SQL queries, statistical visualizations, and interactive maps were used to explore relationships within the data.
- Machine learning models were trained to predict first-stage landing success based on historical mission characteristics.

Objectives

- Collect and prepare Falcon 9 launch data for analysis.
- Explore historical launch records using SQL and data visualization techniques.
- Examine geographic patterns through interactive Folium maps.
- Develop and evaluate supervised machine learning models to predict first-stage landing success.
- Compare model performance using consistent evaluation metrics to assess predictive capability.

Section 1

Methodology

Methodology

Executive Summary

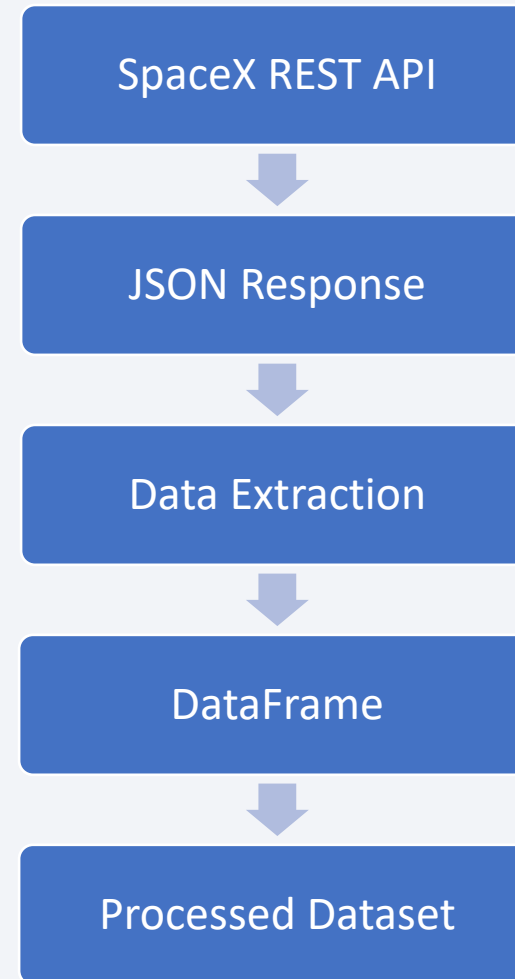
- Data collection methodology:
 - Acquir launch information from public APIs and web sources.
- Perform data wrangling
 - Clean and transform the dataset for analysis
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Train and compare multiple classification models for landing prediction

Data Collection

- Connected to the public SpaceX REST API to retrieve historical Falcon 9 launch data.
- Collected mission details including flight number, launch date, launch site, payload information, booster version, orbit type, and landing outcome.
- Converted the JSON responses into structured data frames for further analysis.
- Extracted additional launch information through web scraping techniques.
- Parsed HTML tables to obtain launch records not directly available through the API.
- Prepared a unified dataset for downstream analytical tasks.

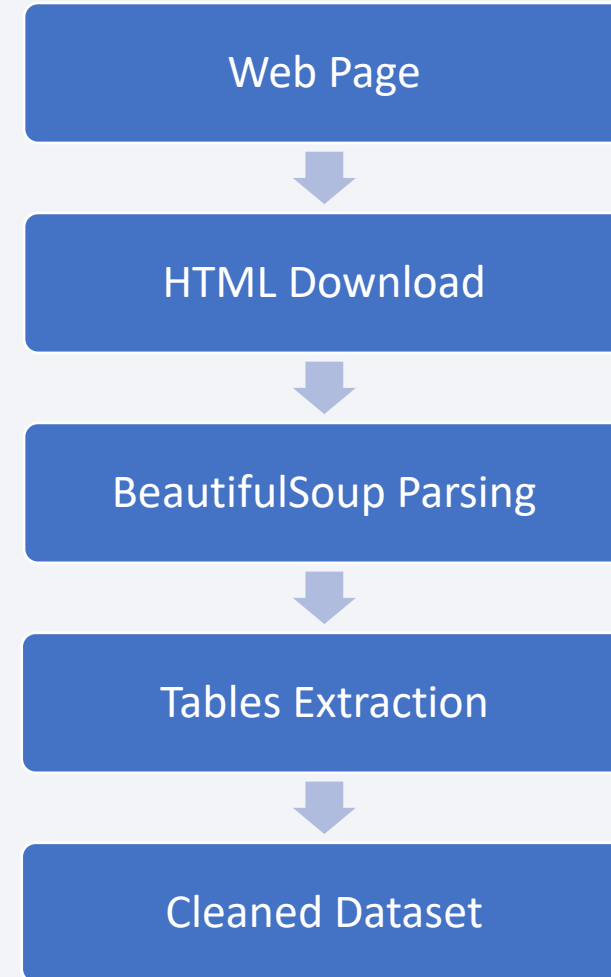
Data Collection – SpaceX API

- Automated data collection directly from the official SpaceX API.
- Retrieved structured mission information with minimal manual processing.
- Created a reproducible pipeline for obtaining launch records.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Data-Collection-API.ipynb>



Data Collection - Scraping

- Supplemented API data using publicly available launch information.
- Applied HTML parsing techniques to extract structured records.
- Improved dataset completeness for subsequent analysis.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Data-Collection-with-Web-Scrapping.ipynb>



Data Wrangling

- Cleaned the dataset by handling missing and inconsistent values.
- Selected relevant mission attributes for subsequent analysis.
- Converted variables into appropriate data types and formats.
- Created a binary landing outcome variable for predictive modeling.
- Produced a structured dataset suitable for SQL analysis, visualization, and machine learning.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Data-Wrangling.ipynb>

EDA with Data Visualization

- Explored the distribution of launch outcomes using statistical visualizations.
- Examined relationships between payload mass and landing success.
- Compared launch performance across launch sites.
- Investigated the influence of orbit type, booster version, and flight number.
- Identified patterns that guided feature selection for predictive modeling.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Complete-the-EDA-with-Visualization.ipynb>

EDA with SQL

- Queried the launch database using SQL to summarize mission characteristics.
- Retrieved launch records by site, booster version, payload, and orbit.
- Calculated summary statistics and mission counts.
- Compared successful and unsuccessful landing outcomes through SQL queries.
- Validated analytical findings obtained from visual exploration.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Complete-the-EDA-with-SQL.ipynb>

Build an Interactive Map with Folium

- Visualized SpaceX launch sites using interactive Folium maps.
- Displayed launch locations with map markers.
- Explored surrounding geographic features and nearby infrastructure.
- Measured spatial relationships between launch sites and nearby points of interest.
- Used geographic visualization to provide additional operational context.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Interactive-Visual-Analytics-with-Folium.ipynb>

Build a Dashboard with Plotly Dash

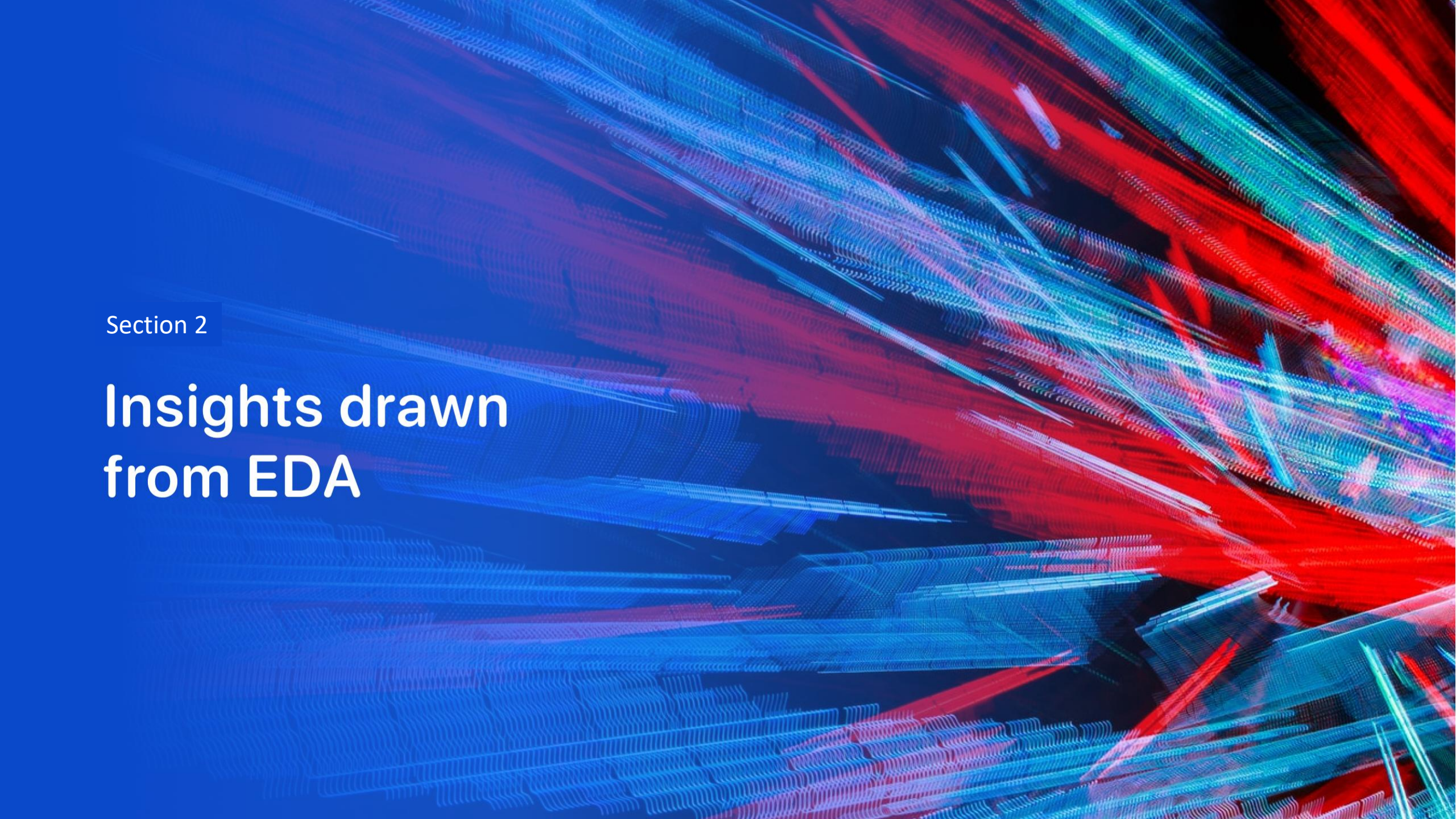
- Developed an interactive Plotly Dash dashboard for exploratory analysis.
- Enabled launch site selection through interactive controls.
- Displayed landing success statistics using dynamic charts.
- Filtered payload ranges to examine mission performance.
- Supported user-driven exploration of launch characteristics and outcomes.
- <https://github.com/farturo93/applied-data-science-capstone/blob/63412f266bea7c91a4e1cb25d35bbb743b726c8d/Build-an-Interactive-Dashboard-with-Plotly-Dash.py>

Predictive Analysis (Classification)

- Selected engineered launch features as model inputs.
- Prepared training and testing datasets for supervised learning.
- Trained multiple classification algorithms to predict landing success.
- Evaluated model performance using standard classification metrics.
- Compared model results and selected the best-performing predictive model.
- <https://github.com/farturo93/applied-data-science-capstone/blob/ef8dfd3a205311800ef58b50f389908b7e769592/Complete-the-Machine-Learning-Prediction.ipynb>

Results

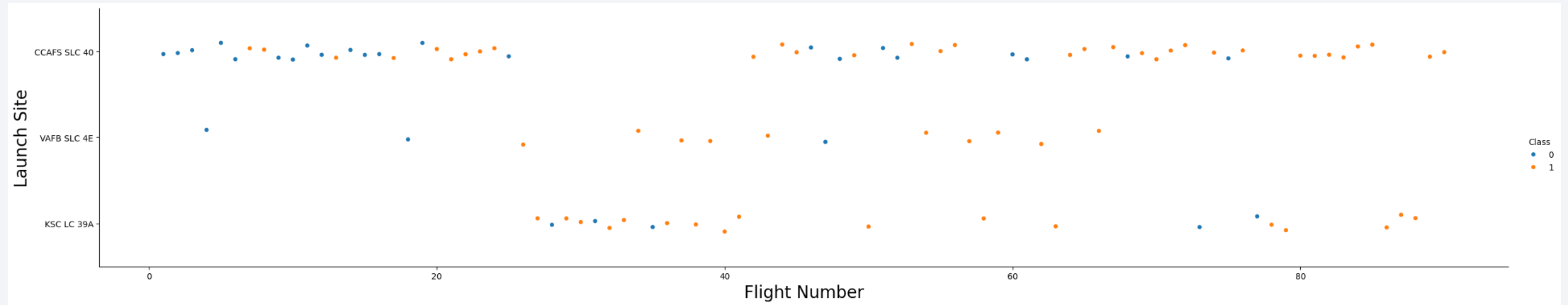
- SQL analysis and visual exploration identified relationships between launch outcomes and mission characteristics.
- Operational variables—including launch site, payload mass, orbit type, booster version, and flight number—were valuable predictors of landing success.
- Multiple classification algorithms were evaluated using standard performance metrics.
- The best-performing model demonstrated strong predictive capability for first-stage landing success.
- Historical launch data provide sufficient information to support data-driven prediction of Falcon 9 landing outcomes.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

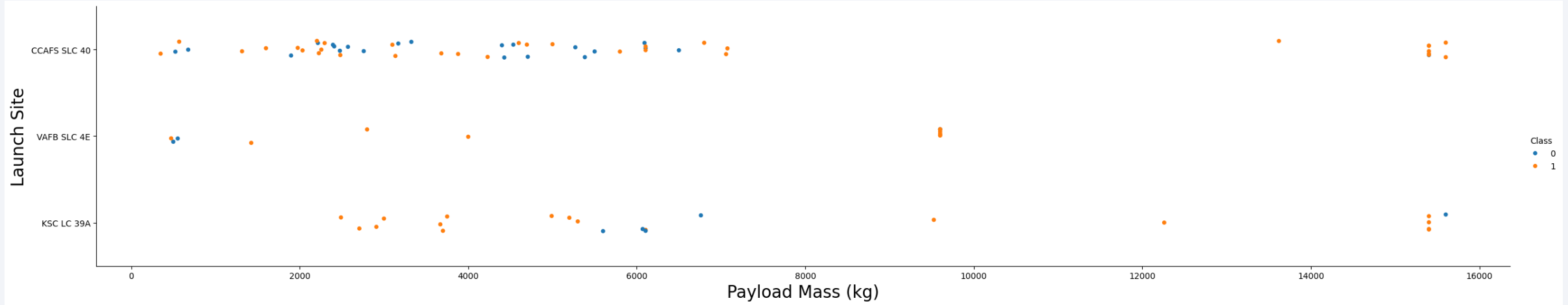
Insights drawn from EDA

Flight Number vs. Launch Site



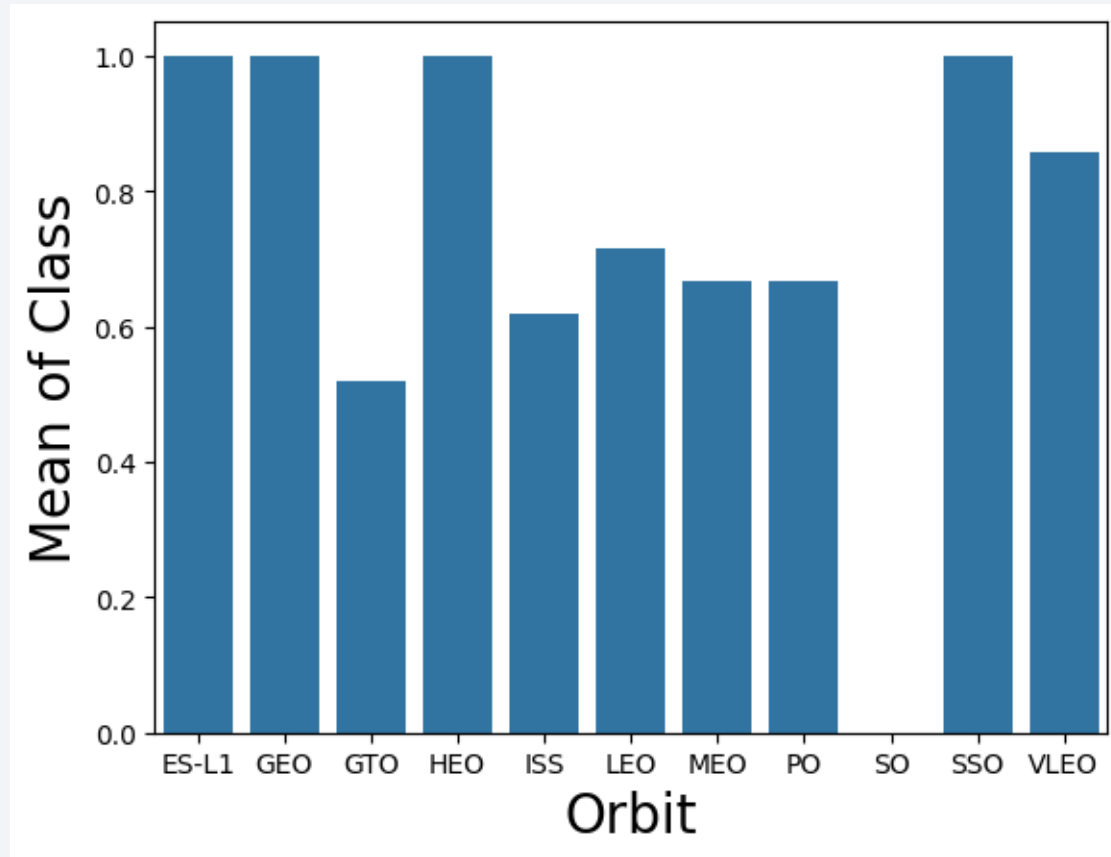
- The scatter plot compares flight number across all launch sites.
- Launch activity increased over time at multiple launch sites.
- Successful launches become more frequent as flight numbers increase.
- The visualization illustrates how launch history evolved across different facilities.

Payload vs. Launch Site



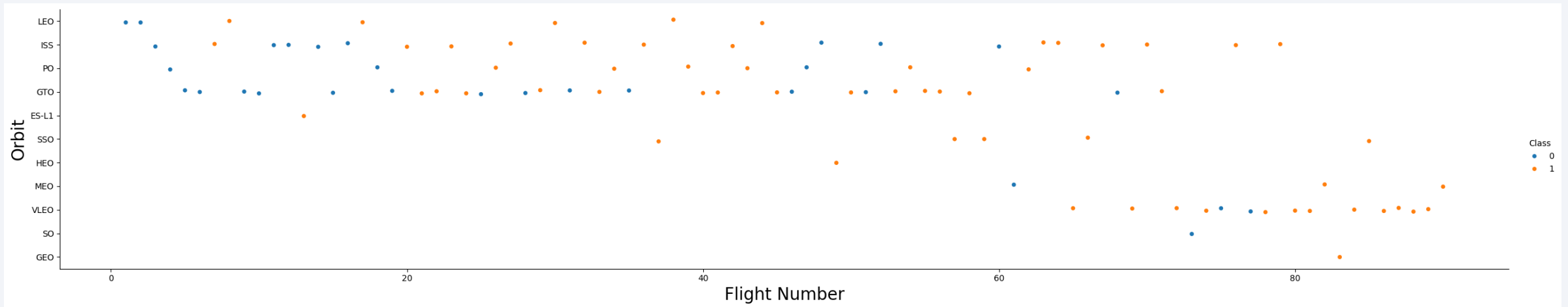
- Payload mass varies across launch sites.
- Both low- and high-payload missions are represented.
- Successful landings occur across a wide payload range.
- Payload mass alone does not fully determine landing success.

Success Rate vs. Orbit Type



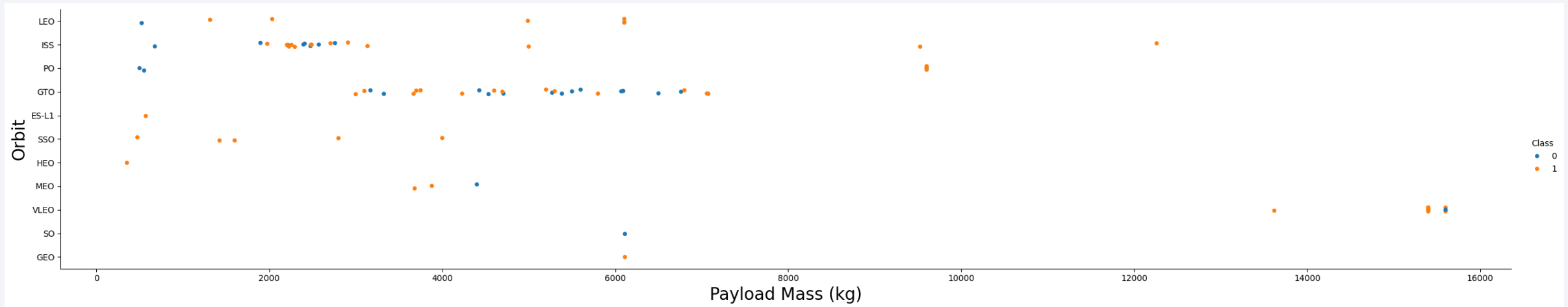
- Landing success varies among orbit categories.
- Some orbit types achieve consistently higher success rates.
- Orbit selection is an important operational variable.
- Orbit type was therefore included as a predictive feature.

Flight Number vs. Orbit Type



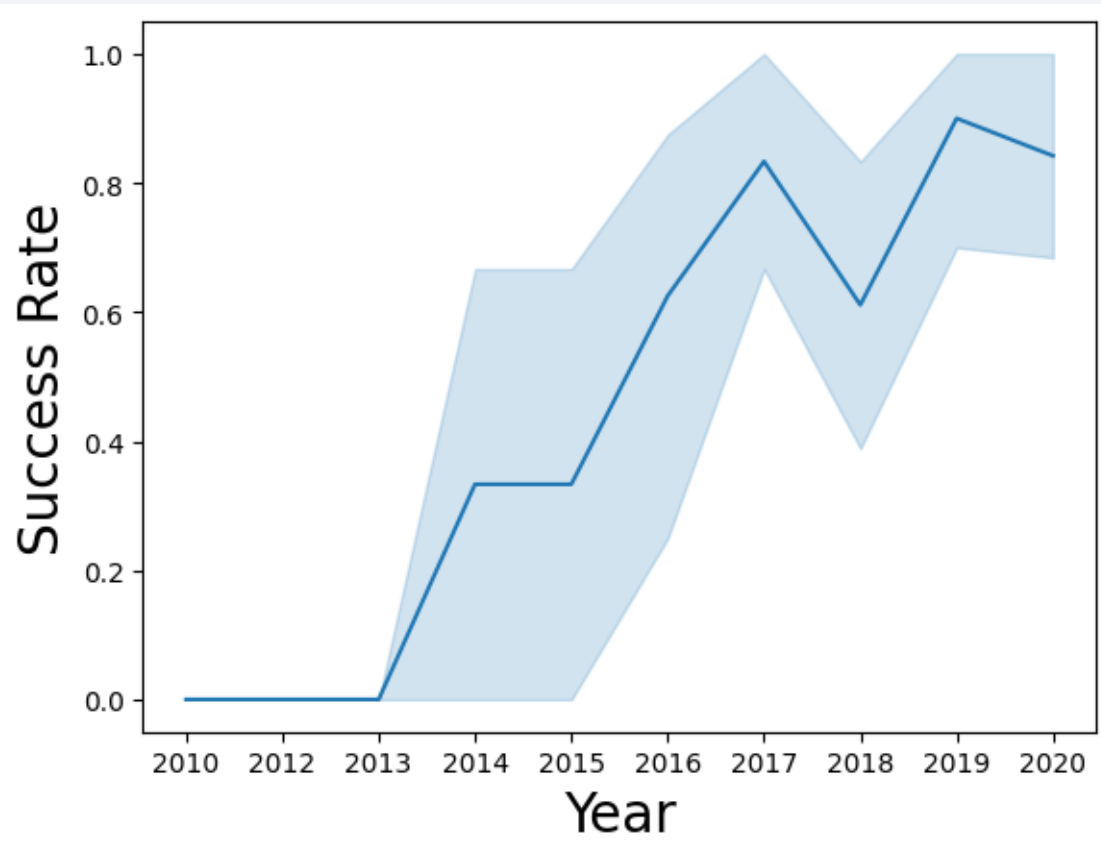
- Later missions are distributed across several orbit types.
- Mission profiles became more diverse over time.
- Landing outcomes improve as flight experience increases.
- Flight number captures operational maturity.

Payload vs. Orbit Type



- Payload distribution differs by orbit.
- Multiple orbit categories include successful landings.
- Payload and orbit jointly influence mission characteristics.
- These variables were retained for machine learning.

Launch Success Yearly Trend



- Annual landing success generally improves over time.
- Later missions demonstrate more consistent successful recoveries.
- The trend reflects increasing operational experience.
- Historical performance supports predictive modeling.

All Launch Site Names

```
%%sql
SELECT DISTINCT "Launch_Site"
FROM SPACEXTBL;

* sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Query returned the unique launch sites contained in the launch dataset.
- Verify available launch locations.
- Support launch-site comparisons throughout the analysis.

Launch Site Names Begin with 'CCA'

```
%%sql
SELECT *
FROM SPACEXTBL
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
```

* [sqlite:///my_data1.db](#)
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Query filtered launch sites beginning with CCA.
- Demonstrate SQL pattern matching.
- Isolate Cape Canaveral launch facilities for analysis.

Total Payload Mass

```
%%sql
SELECT SUM("PAYLOAD_MASS_KG_") AS "Total_NASA_Payload_Mass"
FROM SPACEXTBL
WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my\_data1.db
Done.
```

Total_NASA_Payload_Mass
45596

- Calculated the total payload mass carried for NASA missions.
- Summarize payload delivered by NASA missions.
- Compare mission scale using aggregate statistics.

Average Payload Mass by F9 v1.1

```
%%sql
SELECT AVG("PAYLOAD_MASS_KG_") AS "Average_Payload_Mass"
FROM SPACEXTBL
WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my\_data1.db
Done.
```

Average_Payload_Mass
2928.4

- Calculated the average payload mass for booster version F9 v1.1.
- Compare payload capability by booster version.
- Support engineering performance analysis.

First Successful Ground Landing Date

```
%%sql
SELECT MIN("Date") AS "First_Successful_Ground_Pad_Landing"
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my\_data1.db
Done.
```

First_Successful_Ground_Pad_Landing
2015-12-22

- Retrieved the earliest successful Ground Pad landing.
- Identify a significant milestone in Falcon 9 reusability.
- Demonstrate SQL filtering and chronological queries.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
SELECT "Booster_Version"
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS_KG_" > 4000
  AND "PAYLOAD_MASS_KG_" < 6000;
```

```
* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Listed boosters that:
 - Successfully landed on a drone ship.
 - Carried payloads between 4000 kg and 6000 kg.
- Combine multiple filtering conditions.
- Examine successful medium-payload missions.

Total Number of Successful and Failure Mission Outcomes

```
%%sql
SELECT "Mission_Outcome", COUNT(*) AS "Total_Count"
FROM SPACEXTBL
GROUP BY "Mission_Outcome";
```

* [sqlite:///my_data1.db](#)
Done.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Calculated the total number of successful and failed landing outcomes.
- Summarize landing performance.
- Compare overall mission outcomes.

Boosters Carried Maximum Payload

```
%%sql
SELECT "Booster_Version"
FROM SPACEXTBL
WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);
```

* [sqlite:///my_data1.db](#)
Done.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- Returned the booster(s) carrying the maximum payload.
- Identify missions with the highest payload capacity.
- Demonstrate SQL aggregation using MAX().

2015 Launch Records

```
%%sql
SELECT substr("Date", 6, 2) AS "Month", "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Failure (drone ship)'
| AND substr("Date", 1, 4) = '2015';
```

* [sqlite:///my_data1.db](#)
Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Listed failed drone ship landing missions during 2015.
- Review unsuccessful missions from an early operational period.
- Compare launch site and booster information.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS "Outcome_Count"
FROM SPACEXTBL
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY "Outcome_Count" DESC;
```

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

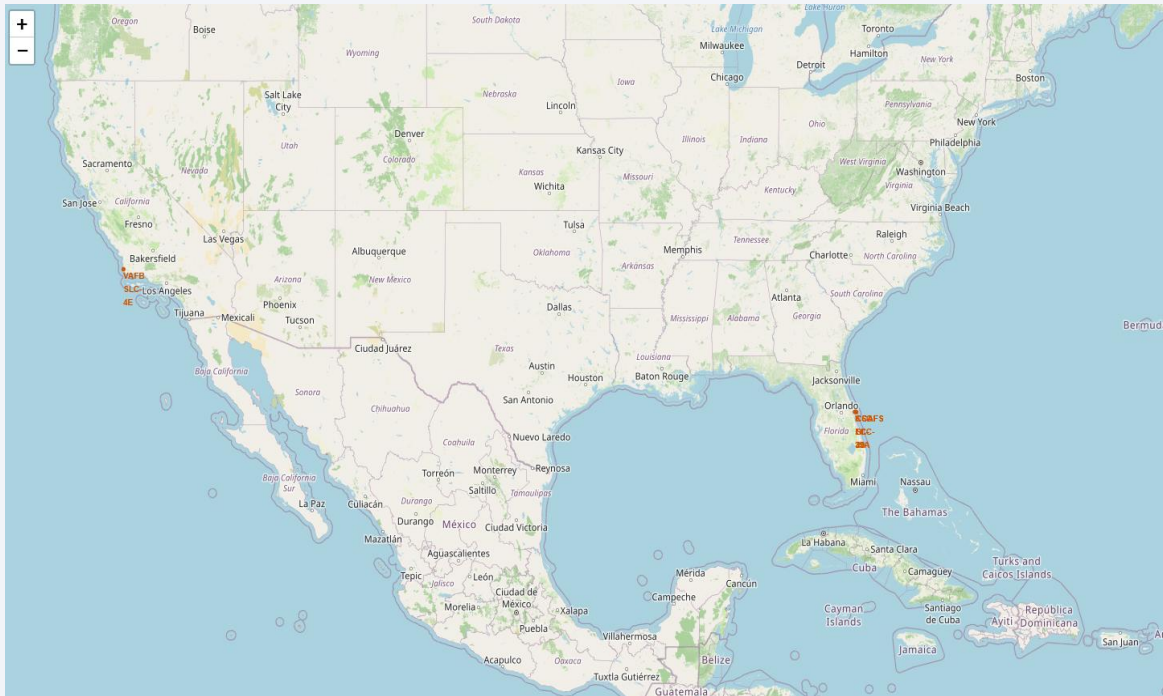
- Ranked landing outcomes between 2010-06-04 and 2017-03-20.
- Compare frequencies of each landing outcome.
- Summarize mission performance over the selected period.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of white clouds and a dense network of yellow and orange lights representing cities and urban areas. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The horizon line is visible, separating the dark sky from the Earth's surface.

Section 3

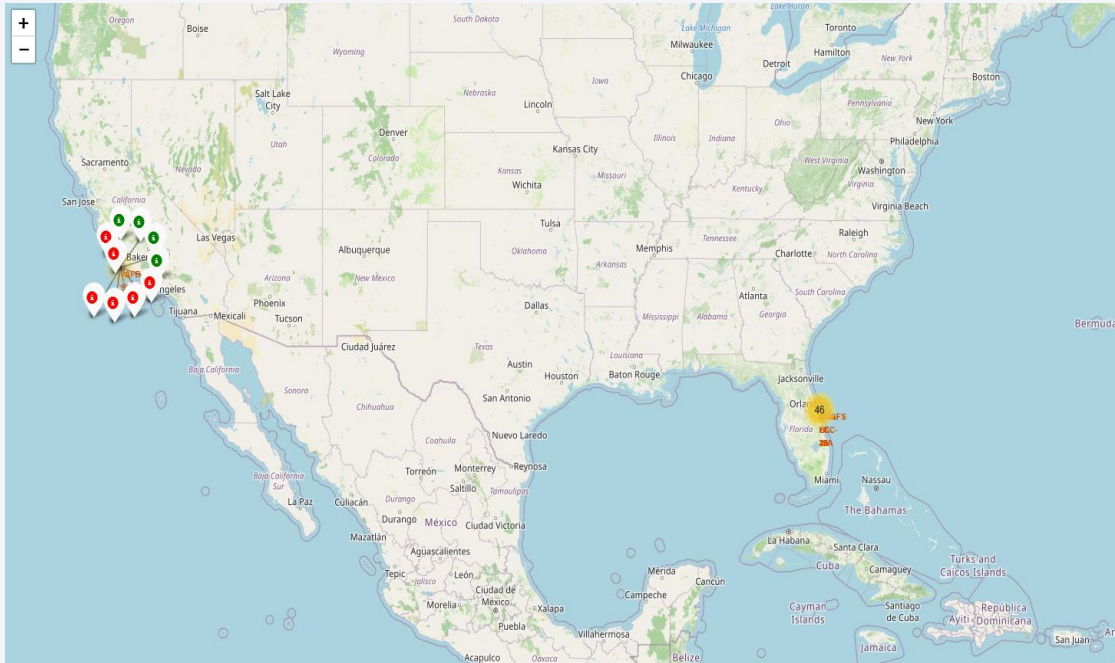
Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites



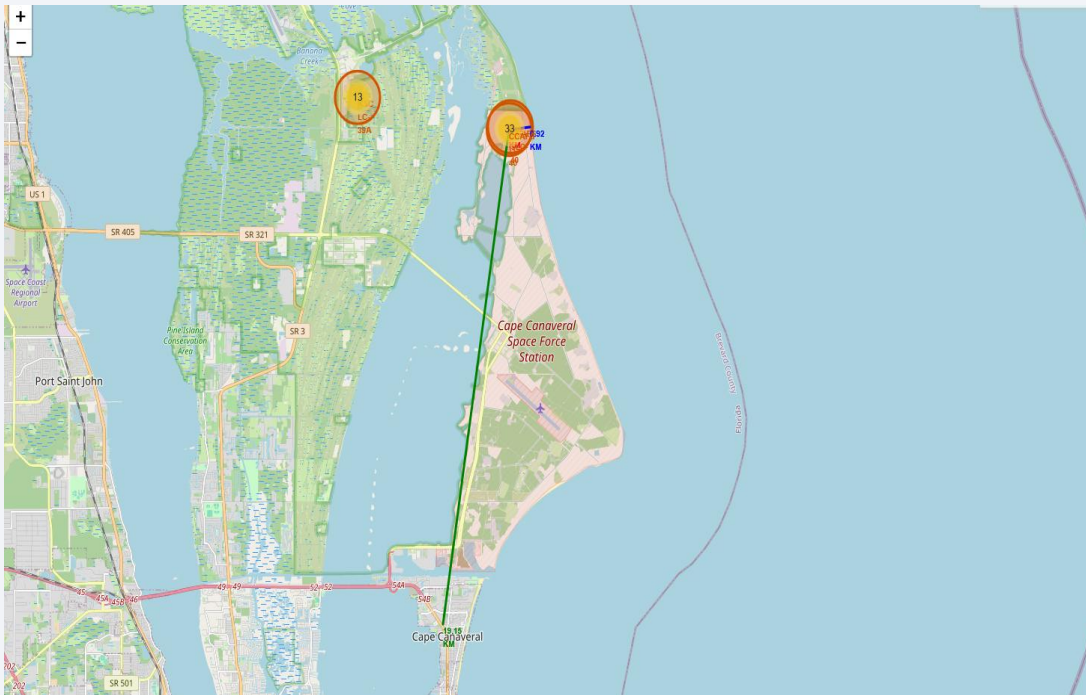
- Interactive Folium map displays all SpaceX launch sites on a geographic map.
- Markers identify the location of each launch facility used in the dataset.
- Geographic visualization provides spatial context for launch operations.
- The map serves as the foundation for further location-based analysis.

Launch Site Landing Outcomes

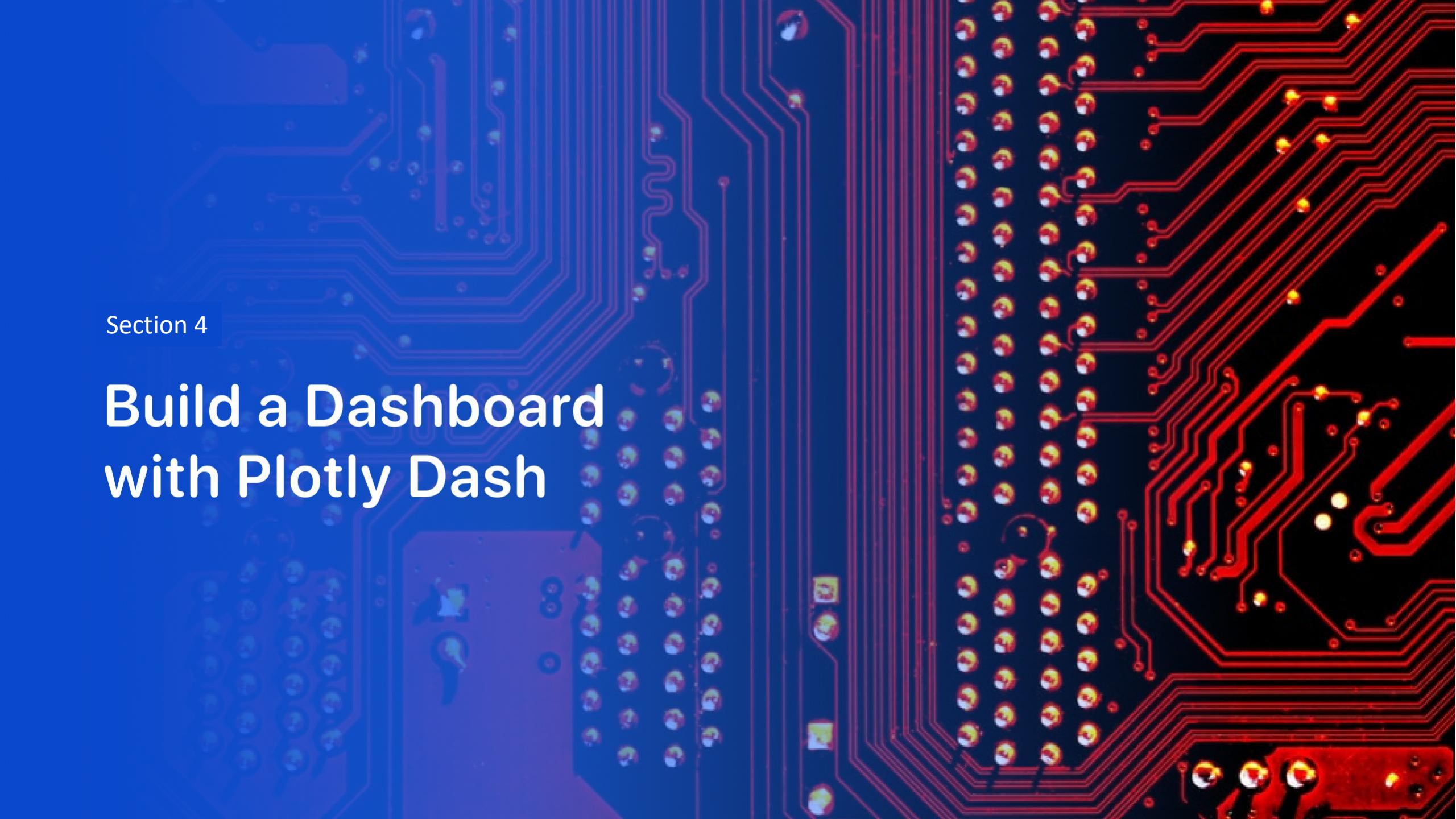


- Landing outcomes are visualized using color-coded map markers.
- Successful and unsuccessful landings can be compared geographically.
- Interactive markers allow exploration of individual launch records.
- Geographic visualization complements the statistical results obtained during EDA.

Launch Site Proximity Analysis



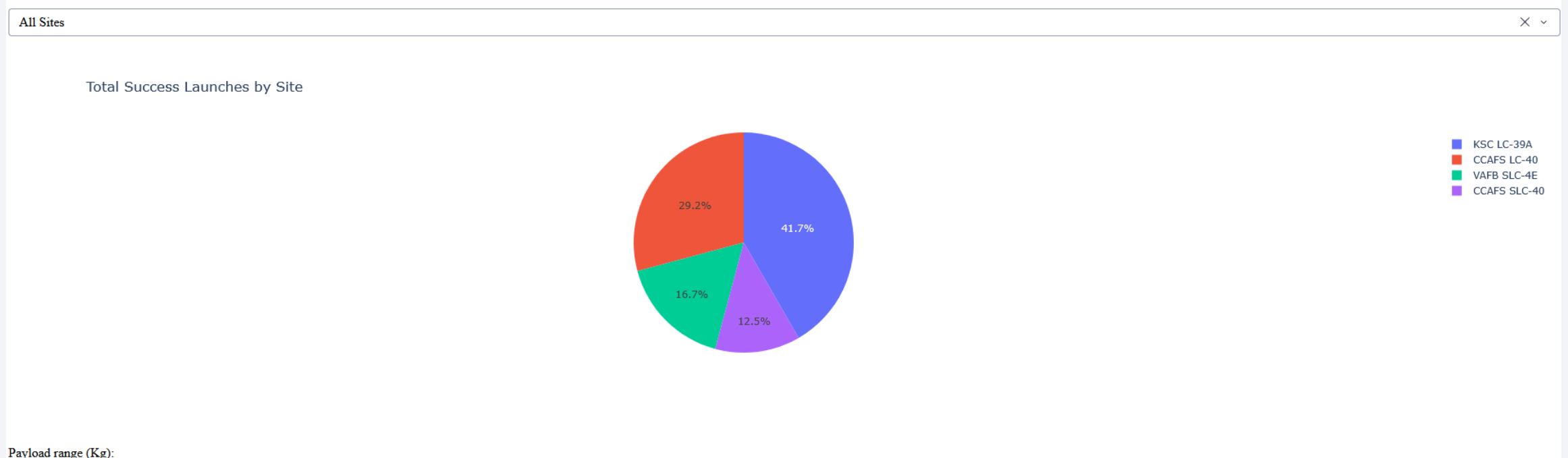
- Distance measurements were calculated from the launch site to nearby infrastructure.
- The map illustrates nearby coastlines, highways, railways, and populated areas.
- Lines and markers highlight important geographic relationships.
- Proximity analysis provides additional operational context for each launch site.



Section 4

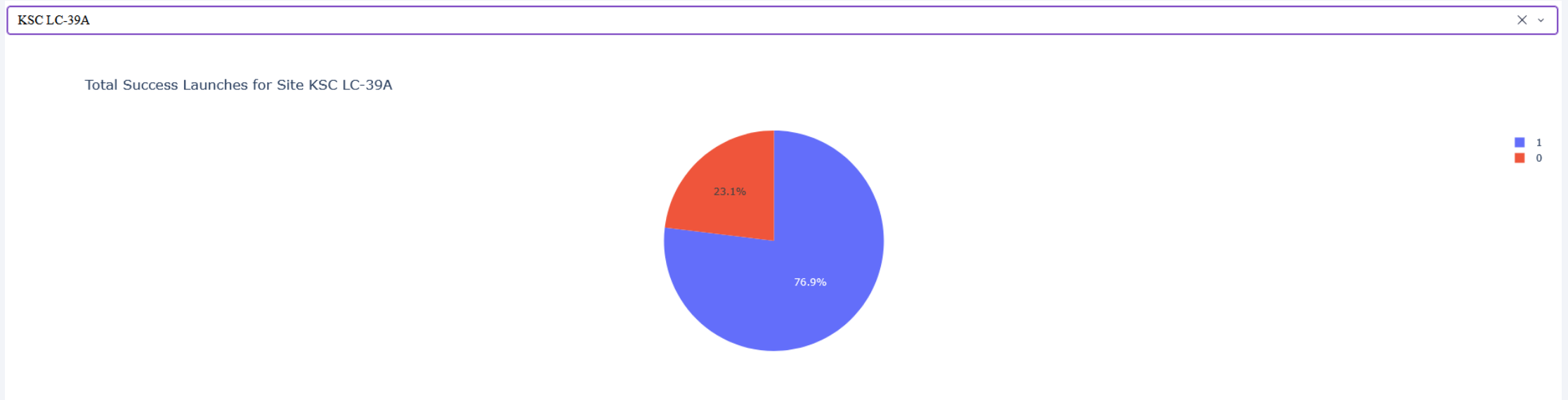
Build a Dashboard with Plotly Dash

Launch Success Count by Site



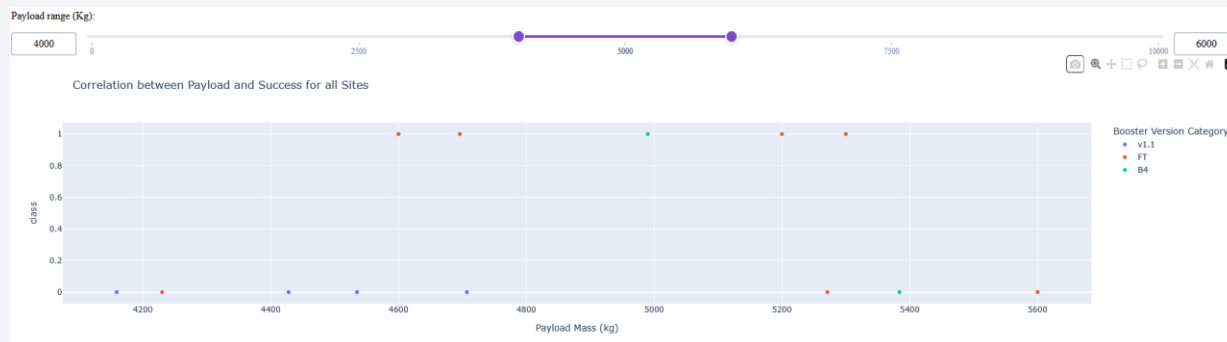
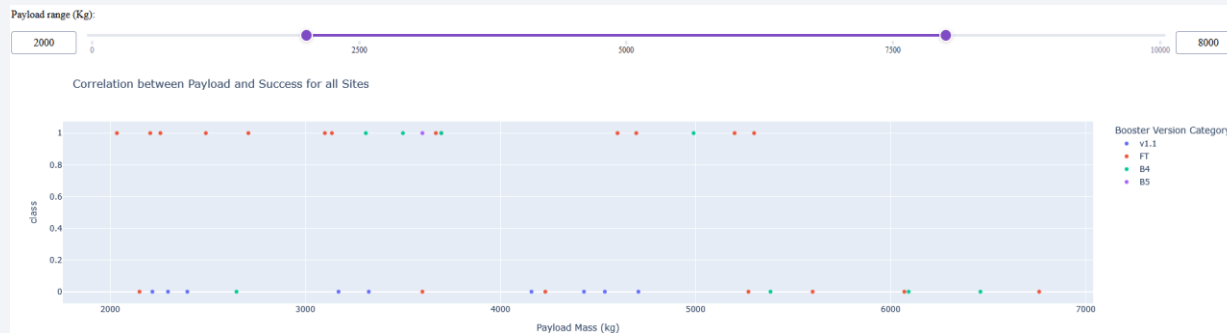
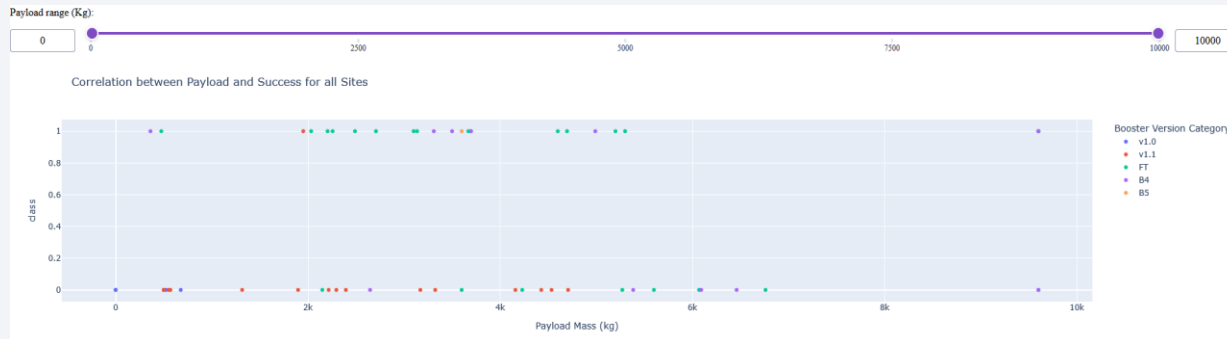
- Interactive pie chart summarizes successful launches across all launch sites.
- Users can compare launch performance between sites.
- Dashboard controls enable dynamic exploration of mission results.
- Provides a high-level overview of launch success distribution.

Launch Success Ratio for a Selected Site



- Selecting an individual launch site updates the pie chart automatically.
- Dashboard interaction highlights the landing success ratio for the selected site.
- Enables comparison of launch performance between individual facilities.
- Demonstrates the value of interactive filtering during exploratory analysis.

Payload Mass vs. Launch Outcome



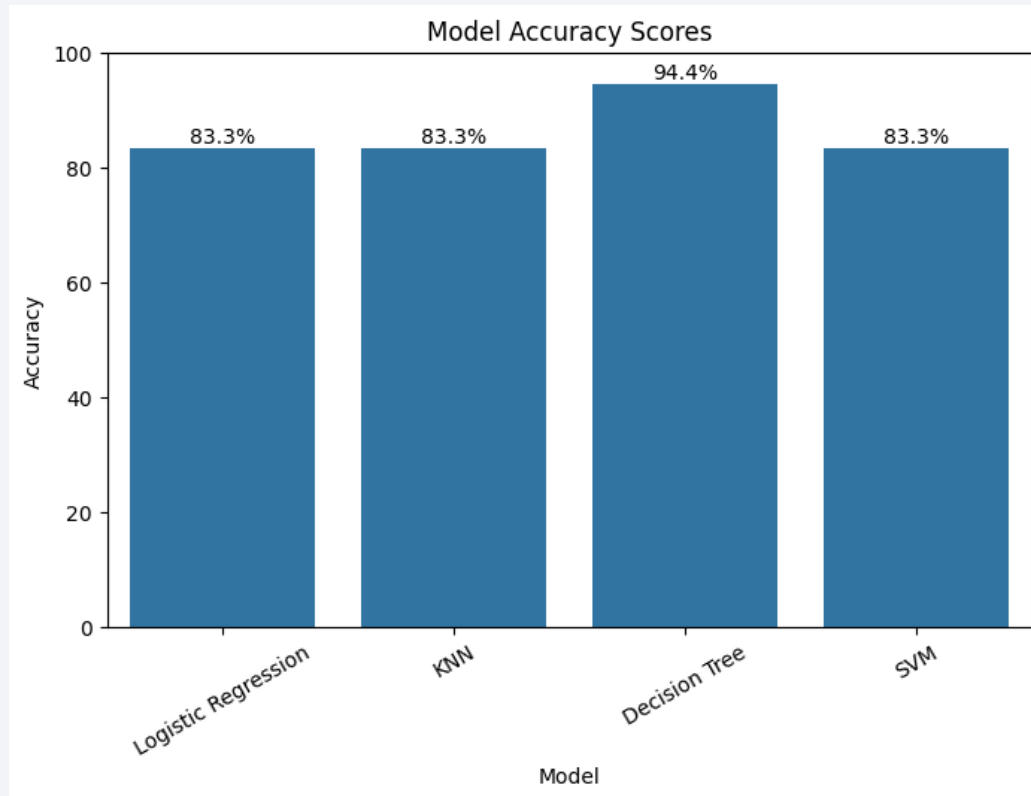
- Scatter plot visualizes the relationship between payload mass and landing outcome.
- Payload range slider filters missions interactively.
- Booster versions are distinguished within the visualization.
- Dashboard interactions allow users to explore how payload characteristics relate to landing success.



Section 5

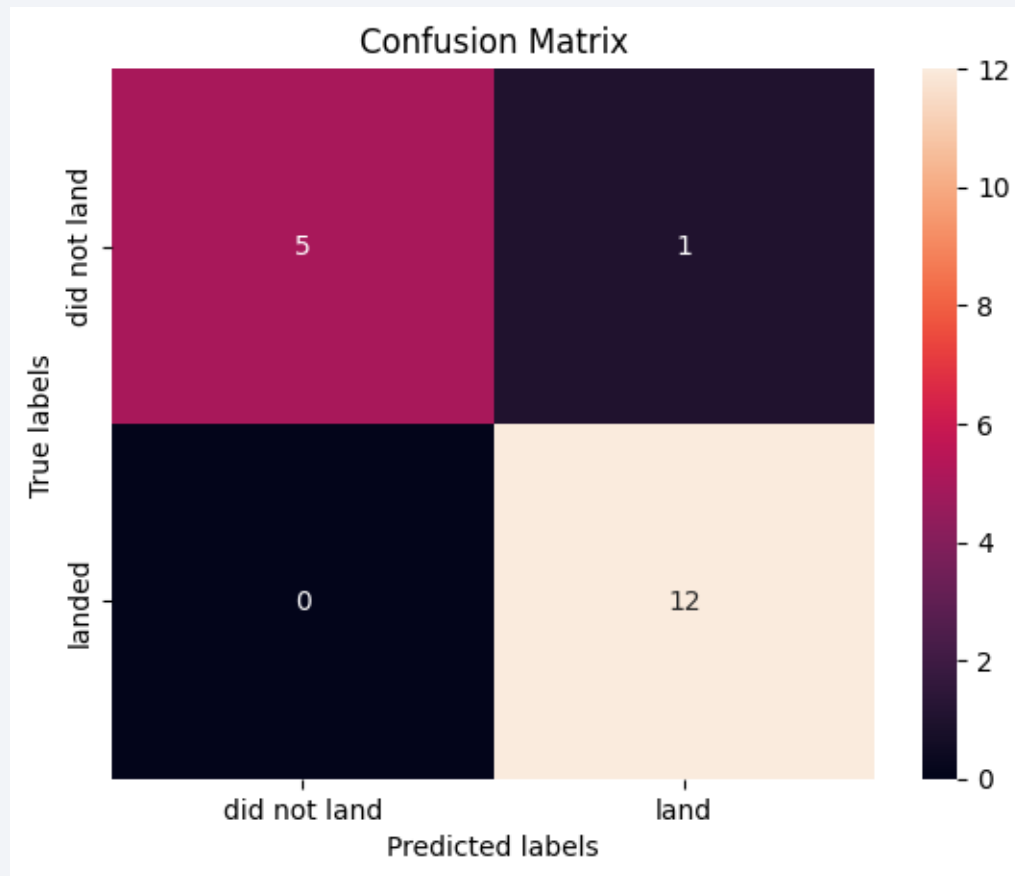
Predictive Analysis (Classification)

Classification Accuracy



- All four classification models were trained using the same processed dataset.
- Hyperparameters were optimized using GridSearchCV with 10-fold cross-validation.
- Each model achieved a test accuracy of 83.3% on the evaluation dataset.
- Since all models produced the same accuracy, no single classifier outperformed the others based on this metric alone.

Confusion Matrix



- The confusion matrix compares predicted landing outcomes with the observed landing outcomes.
- Most observations are correctly classified, indicating good predictive performance.
- Misclassified missions are limited to a relatively small number of cases.
- The confusion matrix complements overall accuracy by illustrating the distribution of correct and incorrect predictions.

Conclusions

- Historical Falcon 9 launch data were successfully collected, cleaned, and analyzed using SQL, statistical visualization, interactive mapping, and machine learning techniques.
- Exploratory analysis identified meaningful relationships between landing success and mission characteristics, including launch site, payload mass, orbit type, booster version, and flight number.
- Four supervised machine learning models (Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors) were trained and evaluated using a consistent workflow with cross-validation and hyperparameter tuning.
- All evaluated classifiers achieved the same test accuracy (**83.3%**), demonstrating that historical launch features provide reliable predictive information for first-stage landing success.

Appendix A – Data Collection

- SpaceX launch dataset retrieved from the public API.
- Falcon 9 launch table extracted through web scraping.
- Initial DataFrame preview used to verify successful data collection.
- Dataset imported into pandas for preprocessing.

Supporting Code

- `requests.get(...)`
- `BeautifulSoup(...)`
- `pd.DataFrame(...)`

Appendix B – SQL Analysis

Representative SQL Queries

- SELECT DISTINCT Launch_Site
- FROM SPACEXTBL;
- SELECT COUNT(*)
- FROM SPACEXTBL
- WHERE CLASS = 1;
- SELECT Booster_Version,
- AVG(PAYLOAD_MASS__KG_)
- FROM SPACEXTBL
- GROUP BY Booster_Version;
- SELECT Orbit,
- COUNT(*)
- FROM SPACEXTBL
- GROUP BY Orbit;

Thank you!

